

# Omar Ayman Zaki

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## SUMMARY

I'm passionate about building backend systems that scale and perform. Across .NET, Java/Spring Boot, and Node.js stacks, I've developed REST APIs, microservices, and distributed systems while applying OOP, SOLID principles, and modern DevOps practices to create production-ready solutions.

## TECHNICAL SKILLS

**Programming Languages:** C#, Java, JavaScript, Python, Dart, PHP, C++, SQL

**Frameworks:** ASP.NET MVC, ASP.NET Core, Spring Boot, Spring Security, Express.js, Node.js, React, Flutter, JavaFX

**Data & Storage:** SQL Server, PostgreSQL, MySQL, MongoDB, SQLite, Entity Framework / EF Core

**DevOps & Infrastructure:** Docker, Apache Kafka, Git, GitHub, JPA, Hibernate, Drift ORM, WebSockets

**Architecture & Patterns:** RESTful APIs, Microservices, OOP, SOLID Principles, Design Patterns, DSA, JWT Auth, MVC, .NET Framework

## WORK EXPERIENCE

**IT Operations Engineer**

May 2025 – Present

e& Egypt (Etisalat Egypt)

Cairo, Egypt

- Maintain IT infrastructure, and ensure high availability of services at one of Egypt's largest telecoms.

## PROJECTS

**Instagram Clone** / C#, ASP.NET MVC, Entity Framework, Bootstrap, SQL Server

- Full-featured photo and video sharing platform with private messaging and user authentication.
- Built with ASP.NET MVC and Entity Framework on a SQL Server backend, with a responsive Bootstrap frontend.

**Compiler Scanner** / C#, ASP.NET, .NET

- Lexical scanner for a case-sensitive OOP language with a .NET-based UI.
- Demonstrates deep familiarity with the .NET platform and language processing concepts.

**Real-Time Chat Application** / Java, Spring Boot, WebSockets, JWT, Spring Security

- Secure real-time messaging using WebSockets and JWT/Spring Security authentication.
- Architecture mirrors ASP.NET Core + SignalR patterns; easily portable to the .NET stack.

**E-Commerce REST API** / Java, Spring Boot, PostgreSQL, JPA/Hibernate

- Full REST API backend handling products, carts, and orders via JPA/Hibernate.
- Comparable in design to an ASP.NET Core Web API project with Entity Framework Core.

**Incident Report Tool** / Java, Spring Boot, REST API

- A lightweight Spring Boot tool for logging, managing, and exporting structured shift incident reports.
- Features REST APIs, JSON file storage, CSV import, SMS notifications, and audit trails.

**Microservices Architecture App** / Node.js, Express.js, Apache Kafka, Docker

- Async inter-service communication via Apache Kafka, fully containerized with Docker.

**Lift Tracker** / Flutter, Dart, SQLite, Drift ORM

- Cross-platform workout tracking app with an offline-first SQLite database via Drift ORM.

**Stereo R-CNN 3D Object Detection** / Python, Deep Learning, Computer Vision

- LiDAR-free 3D object detection system for autonomous driving using stereo cameras.
- Validated on the KITTI benchmark with competitive accuracy.

**Job Board REST API** / C#, ASP.NET Core Web API, Entity Framework Core, SQL Server, JWT

- Built a fully documented REST API for job listings, applications, and employer/candidate profiles.
- Implemented JWT-based authentication with role-based authorization (Admin, Employer, Candidate) using ASP.NET Core Identity.
- Used EF Core with SQL Server, repository pattern, and pagination for scalable data access.

**Fingerprint Recognition System** / Python, ANN

- ANN-based biometric identification analyzing ridge endings and bifurcations.

**Course Management System** / Java, JavaFX

- Desktop MVC app for managing courses, students, and assignments.

## EDUCATION

**Bachelor of Science in Computer Science**

Aug 2019 – Jun 2023

Helwan University, Faculty of Computer and Artificial Intelligence

GPA: 3.4 / 4.0

## LANGUAGES

**Arabic:** Native | **English:** Intermediate

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